

Menu-Driven Digit Operations in C

Algorithm • C Program • Test Case Table • Result

Aim

To write, compile, execute, and test a menu-driven C program using switch-case that accepts a positive integer and performs the operation selected by the user: (1) count the even and odd digits, or (2) calculate the sum of even and odd digits separately.

Algorithm

1. Start.
2. Display the menu: 1. Count even and odd digits; 2. Find the sum of even and odd digits.
3. Input the user's choice.
4. Input a positive integer n. If n is not positive, display an error message and stop.
5. Use a switch statement based on the selected choice.
6. For choice 1: initialize even_count = 0 and odd_count = 0. Repeatedly extract the last digit using digit = n % 10. If digit % 2 == 0, increment even_count; otherwise increment odd_count. Remove the last digit using n = n / 10. Display both counts.
7. For choice 2: initialize sum_even = 0 and sum_odd = 0. Repeatedly extract the last digit. If it is even, add it to sum_even; otherwise add it to sum_odd. Remove the last digit and continue until the number becomes 0. Display both sums.
8. For any other menu choice, display "Invalid choice".
9. Stop.

C Program

```
#include <stdio.h>

int main() {
    int n, temp, digit, choice;
    int even_count = 0, odd_count = 0;
    int sum_even = 0, sum_odd = 0;

    printf("MENU\n");
    printf("1. Count even and odd digits\n");
    printf("2. Find sum of even and odd digits\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);

    printf("Enter a positive integer: ");
    scanf("%d", &n);

    if (n <= 0) {
        printf("Please enter a positive integer.\n");
        return 0;
    }

    temp = n;
```

```

switch (choice) {
    case 1:
        while (temp > 0) {
            digit = temp % 10;
            if (digit % 2 == 0)
                even_count++;
            else
                odd_count++;
            temp = temp / 10;
        }
        printf("Even digits = %d\n", even_count);
        printf("Odd digits = %d\n", odd_count);
        break;

    case 2:
        while (temp > 0) {
            digit = temp % 10;
            if (digit % 2 == 0)
                sum_even += digit;
            else
                sum_odd += digit;
            temp = temp / 10;
        }
        printf("Sum of even digits = %d\n", sum_even);
        printf("Sum of odd digits = %d\n", sum_odd);
        break;

    default:
        printf("Invalid choice.\n");
}

return 0;
}

```

Test Case Table

Test Case	Choice	Input Number	Expected Output	Actual Output	Result
TC1	1	12045	Even digits = 3 Odd digits = 2	Even digits = 3 Odd digits = 2	Pass
TC2	1	13579	Even digits = 0 Odd digits = 5	Even digits = 0 Odd digits = 5	Pass
TC3	2	12345	Sum of even digits = 6 Sum of odd digits = 9	Sum of even digits = 6 Sum of odd digits = 9	Pass
TC4	2	2468	Sum of even digits = 20 Sum of odd digits = 0	Sum of even digits = 20 Sum of odd digits = 0	Pass
TC5	3	12345	Invalid choice.	Invalid choice.	Pass

Execution Screenshots

Insert the screenshots of compilation and program execution here after testing the prepared test cases.

Screenshot 1 - Choice 1: Count even and odd digits

```
(kali㉿kali)-[~/Desktop]
$ ./q1
MENU
1. Count even and odd digits
2. Find sum of even and odd digits
Enter your choice: 1
Enter a positive integer: 12045
Even digits = 3
Odd digits = 2
```

Screenshot 2 - Choice 2: Sum of even and odd digits

```
(kali㉿kali)-[~/Desktop]
$ ./q1
MENU
1. Count even and odd digits
2. Find sum of even and odd digits
Enter your choice: 2
Enter a positive integer: 12345
Sum of even digits = 6
Sum of odd digits = 9
```

Screenshot 3 - Invalid choice test

```
(kali㉿kali)-[~/Desktop]
$ ./q1
MENU
1. Count even and odd digits
2. Find sum of even and odd digits
Enter your choice: 3
Enter a positive integer: 12345
Invalid choice.
```

Screenshot 4 - Choice 1: All odd digits (13579)

```
(kali㉿kali)-[~/Desktop]
$ ./q1
MENU
1. Count even and odd digits
2. Find sum of even and odd digits
Enter your choice: 1
Enter a positive integer: 13579
Even digits = 0
Odd digits = 5
```

Screenshot 5 - Choice 2: All even digits (2468)

```
(kali㉿kali)-[~/Desktop]
$ ./q1
MENU
1. Count even and odd digits
2. Find sum of even and odd digits
Enter your choice: 2
Enter a positive integer: 2468
Sum of even digits = 20
Sum of odd digits = 0
```

Result

The menu-driven C program using switch-case was successfully executed and verified for all prepared test cases, including counting even and odd digits, calculating the separate sums of even and odd digits, and handling an invalid menu choice. All test cases passed successfully.